

COUNTEREXAMPLES TO THE MONOTONE COMPLETENESS CONJECTURE

JANANAN ARULSEELAN AND JAMES E. HANSON

ABSTRACT. Combining a modification of the recent counterexample to the quasitrace conjecture with model-theoretic reductions due to the first author, we exhibit a counterexample to the monotone completeness conjecture. We fully reproduce the disproof of the quasitrace conjecture by a modified construction and arguments.

1. INTRODUCTION

The two major open problems in the theory of AW^* -algebras were the Kaplansky conjecture and the monotone completeness conjecture. Recently, an AI system from OpenAI claimed to produce a counterexample to the former in an unpublished manuscript (September 2026) which we henceforth refer to as "the quasitrace manuscript". We give a modification of that construction which gives uniform failure of traciality. Using this and model theoretic results of the first author, we produce a counterexample to the monotone completeness conjecture. Our construction does not appear suited to properly infinite AW^* -algebras, so that case of the monotone completeness conjecture remains open. Christensen and Pedersen show that all properly infinite AW^* -algebras are monotone sequentially complete. Our negative resolution here stands in an interesting contrast with our recent positive resolution of the weaker normality conjecture [2].

Note that failure of monotone completeness does not immediately follow from the abstract failure of the quasitrace conjecture, so there is genuine work to do. The bulk of our work lies in the following theorem asserting the existence of a finite AW^* -factor admitting a certain configuration of elements. Namely,

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Theorem. *There is a finite AW*-factor M with canonical normalized quasitrace τ and elements $y_1, y_2, y_3 \in M$ such that*

$$\sum_{i=1}^3 y_i^* y_i = 1, \quad \sum_{i=1}^3 y_i y_i^* \leq \frac{1}{2} 1.$$

This configuration is the new ingredient. It yields positive contractions $a \leq b$ satisfying

$$\tau(a) = \tau(b) = \frac{1}{2}, \quad \tau(b - a) = \frac{1}{2}.$$

Equal quasitrace and order force a and b to have the same scalar spectral distribution. They are therefore approximately unitarily conjugate, and become exactly conjugate in a quasitracial ultrapower. Iterating the conjugating unitary gives an increasing orbit with a fixed quasitracial gap.

Theorem. *There is a type II_1 AW*-factor which is not monotone sequentially complete. Moreover, there is such a factor which is d_τ -separable.*

Thus the monotone-completeness conjecture fails. The construction also answers Kaplansky's question before the ultrapower is taken. That is, our construction builds finitely generated unital C*-algebras each with a normalized 2-quasitrace which is not a trace, and therefore type II_1 AW*-factors which are not von Neumann algebras (Proposition 5.5).

These are Theorems A and B respectively. Section 2 gives the model theoretic reduction from our configuration in the former to failure of monotone completeness as summarized above.

The construction involved in Theorem A has to balance three competing requirements, explained below, and takes up Sections 3, 4, 5, and 6. The algebra in Theorem A is a quasitracial ultrapower of a family of AW*-factors obtained from finitely generated algebras B_N , which in turn come from concrete algebras of operators on Hilbert bundles coming from finite-rank complex bundles over products of complex projective spaces.

The B_N s are arranged to satisfy

$$\sum_k s_k^* s_k = 1, \quad \sum_k s_k s_k^* \leq \gamma_N 1,$$

with $\gamma_N \rightarrow 1/2$ as $N \rightarrow \infty$. We arrange this via compression and normalization arguments that produce families of bundle maps whose

ranges collectively expand every fiber subspace by a factor approaching two, and then normalize these maps to obtain the desired operator relations. The difficulty is to achieve these relations while ensuring that the B_N admit quasitraces. For this, the bundles are chosen to have nonvanishing Chern classes which obstruct the existence of certain large trivial subbundles. These obstructions prevent the compression relations from being upgraded to witnesses of proper infiniteness. Section 3 develops the required general results on bundle maps and Section 4 combines them with the topological obstructions to construct the finite bundle systems.

In Section 5, we assemble these finite systems into the algebras B_N . The construction incorporates tests for every matrix size and every possible bound on the word length of a polynomial in the generators. Since the generators have propagation at most one between successive bundle levels, any hypothetical witness of proper infiniteness would yield a forbidden trivial subbundle in one of these tests. Thus no matrix algebra over B_N is properly infinite, and the Cuntz–Blackadar–Handelman theorem gives a normalized quasitrace. Taking an extremal quasitrace and its Haagerup completion then produces a finite AW^* -factor.

The final difficulty is that the preceding construction requires more than the three operators appearing in the theorem. To deal with this, the bundle maps are divided into three maps taking values in summands inherited from the preceding level and eight auxiliary maps taking values in newly introduced summands. The auxiliary maps are needed to obtain the required expansion, but the construction arranges that their ranges lie in a projection of quasitrace $1/N$, even though the corresponding vector bundles have large ranks. Consequently, the auxiliary maps tend to zero in the quasitracial metric. Section 6 takes the quasitracial ultraproduct as $N \rightarrow \infty$, in which these maps vanish and the remaining three operators satisfy the exact relations of Theorem A.

Further consequences are established in Section 7. By the Haagerup characterization [13, Theorem 3.1], a unital C^* -algebra containing $x_1, \dots, x_n$ with

$$\sum_{j=1}^n x_j^* x_j = 1, \quad \left\| \sum_{j=1}^n x_j x_j^* \right\| < 1$$

has no tracial state. Milhøj and Rørdam proposed looking for a counterexample to Kaplansky’s question as an ultraproduct of algebras in which proper infiniteness of matrix algebras appears only at larger and

larger sizes, while the number of elements needed for relations of this form stays bounded [13, Remark 4.7]. The algebras B_N below realize this mechanism with no properly infinite matrix algebra at all, so our modified counterexample to the quasitrace conjecture needs no ultraproduct; see Remark 5.6.

Model theory enters our argument at three points, all through [1]. First, the eight fresh operators in M_N are small in the metric d_τ , so they vanish in the quasitracial ultraproduct. Second, countable saturation of quasitracial ultrapowers turns approximate unitary conjugacy into exact conjugacy and shows that the resulting increasing sequence has no supremum. Third, separable reflection [1, Proposition 5.1 and Theorem 4.12] passes from the nonseparable ultrapower to a d_τ -separable counterexample. The first is dispensable and could be done operator-algebraically. The second and third seem to be essential.

AI disclosure. The idea for the proof was developed by the authors. It was then completed with a preliminary write-up given by GPT 6 Astra. The arguments were significantly reworked and fixed by the authors. The authors wrote the paper and are responsible for its contents. The bundle construction here adapts a construction from an unpublished manuscript produced by an AI system from OpenAI. Every step used here is proved in the present paper or taken from the works listed below.

2. MODEL THEORETIC REDUCTIONS

Throughout this section M is a finite AW*-factor with canonical normalized quasitrace τ . We use the following facts; see [12, 3, 10, 1] (see also [1, Fact 2.5]).

Fact 2.1. *The quasitrace τ is order preserving and linear on abelian C^* -subalgebras of M , and*

$$\tau(x^*x) = \tau(xx^*) \quad (x \in M).$$

Fact 2.2. *$\text{Proj}(M)$ is a complete lattice, and Kaplansky's parallelogram law*

$$p - p \wedge q \sim p \vee q - q$$

holds for $p, q \in \text{Proj}(M)$. On $\text{Proj}(M)$, τ is the normalized dimension function of M : it is faithful and additive on orthogonal projections, and $p \sim q$ if and only if $\tau(p) = \tau(q)$.

Fact 2.3. *Every maximal abelian self-adjoint subalgebra C of M is an AW^* -subalgebra, so $C \cong C(\Omega)$ with Ω extremally disconnected. The restriction of τ to C is a normal state.*

We now begin to show that the configuration of Theorem A yields failure of monotone completeness. The next lemma shows that such a configuration is seen by the quasitrace.

Lemma 2.4. *Suppose that M contains y_1, y_2, y_3 such that*

$$\sum_{i=1}^3 y_i^* y_i = 1, \quad \sum_{i=1}^3 y_i y_i^* \leq \frac{1}{2} 1.$$

Set

$$a = 1 - 2y_1^* y_1, \quad b = 2y_3^* y_3.$$

Then a, b are positive contractions and

$$a \leq b, \quad \tau(a) = \tau(b) = \frac{1}{2}, \quad \tau(b - a) = \frac{1}{2}.$$

Proof. Put

$$u_i = y_i^* y_i, \quad v_i = y_i y_i^*, \quad \alpha_i = \tau(u_i) = \tau(v_i).$$

For $i \neq j$ we have

$$v_j \leq \frac{1}{2} 1 - v_i.$$

The right-hand side belongs to $C^*(1, v_i)$, so order preservation and linearity on this abelian algebra give

$$\alpha_j \leq \frac{1}{2} - \alpha_i.$$

Thus

$$\alpha_i + \alpha_j \leq \frac{1}{2}.$$

Also $v_i \leq \frac{1}{2} 1$, hence $\|u_i\| = \|v_i\| \leq 1/2$. Set

$$A_i = 1 - 2u_i.$$

Then $A_i \geq 0$ and, because $\sum_i u_i = 1$,

$$A_1 + A_2 + A_3 = 1.$$

The inequality $A_j \leq 1 - A_i$ and linearity on $C^*(1, A_i)$ give

$$\tau(A_j) \leq 1 - \tau(A_i),$$

that is,

$$\alpha_i + \alpha_j \geq \frac{1}{2}.$$

Hence every pair sum is $1/2$, and therefore

$$\alpha_1 = \alpha_2 = \alpha_3 = \frac{1}{4}.$$

It follows that $\tau(A_i) = 1/2$ for every i . Finally,

$$a = A_1, \quad b = A_1 + A_2 = 1 - A_3, \quad b - a = A_2,$$

which proves the assertions. $\square$

We next record the order-theoretic consequence.

Let $0 \leq a \leq 1$ in M , and fix a maximal abelian self-adjoint subalgebra $C \cong C(\Omega)$ containing a . For $t \in \mathbb{R}$ let $e_t(a) \in C$ be the characteristic function of the clopen set

$$\overline{\{\omega \in \Omega : a(\omega) > t\}},$$

the closure of $\{a > t\}$; equivalently, $e_t(a)$ is the support projection of $(a - t)_+$. Then $e_t(a)$ commutes with a , and

$$(1) \quad ae_t(a) \geq te_t(a), \quad a(1 - e_t(a)) \leq t(1 - e_t(a)).$$

Let μ_a be the probability measure on $[0, 1]$ with $\tau(f(a)) = \int f d\mu_a$ for $f \in C([0, 1])$. The functions $\min\{1, n(s - t)_+\}$ increase to the indicator function of (t, ∞) , and their values at a increase to $e_t(a)$ in C . Fact 2.3 and monotone convergence therefore give

$$F_a(t) := \tau(e_t(a)) = \mu_a((t, 1]).$$

In particular, F_a is right-continuous.

We use the following facts from [1].

Lemma 2.5. [1, Lemma 4.1] *Let $0 \leq a \leq b \leq 1$ in M . Then*

$$\tau(f(a)) \leq \tau(f(b))$$

for every nondecreasing $f \in C([0, 1])$.

Fact 2.6. *Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. The quasitracial ultrapower $M^{\mathcal{U}}$ is a finite AW*-factor [1, Corollary 4.13], and it is countably saturated. For self-adjoint x, y the relation $x \leq y$ is expressed by the closed condition $\tau((x - y)_+) = 0$ (see [1, Lemma 4.10]).*

Proposition 2.7. *Suppose M contains positive contractions $0 \leq a \leq b \leq 1$ with*

$$\tau(a) = \tau(b), \quad \tau(b - a) = \eta > 0.$$

Then a quasitracial ultrapower of M contains an increasing sequence (c_n) of positive contractions such that for all $m > n$

$$\tau(c_m - c_n) \geq \eta.$$

This sequence has no supremum.

Proof. We first show that a and b have the same scalar spectral distribution. Fix $t \in [0, 1]$ and put

$$f_t(s) = (s - t)_+, \quad g_t(s) = \min\{s, t\} \quad (0 \leq s \leq 1).$$

Both functions are continuous and nondecreasing, so Lemma 2.5 gives

$$\tau(f_t(a)) \leq \tau(f_t(b)), \quad \tau(g_t(a)) \leq \tau(g_t(b)).$$

Since $f_t + g_t$ is the identity function on $[0, 1]$ and $\tau(a) = \tau(b)$, both inequalities are equalities. The functions 1, s , and $(s - t)_+$ span the continuous piecewise linear functions, which are uniformly dense in $C([0, 1])$. Hence

$$\tau(f(a)) = \tau(f(b))$$

for every $f \in C([0, 1])$, so $\mu_a = \mu_b$ and $F_a = F_b$. Thus corresponding spectral projections of a and b have equal quasitraces.

Take spectral partitions of $[0, 1]$ with mesh $1/n$. Corresponding spectral projections have the same quasitrace and hence are Murray–von Neumann equivalent in the finite factor. Summing implementing partial isometries gives a unitary $w_n \in M$ such that

$$\|w_n a w_n^* - b\| \leq \frac{2}{n}.$$

Let $P = M^{\mathcal{U}}$ be a quasitracial ultrapower over a nonprincipal ultrafilter on $\mathbb{N}$. Then

$$w = [w_n] \in P$$

is unitary and

$$w a w^* = b.$$

Set

$$c_n = w^n a w^{-n}.$$

Since $a \leq b = w a w^*$,

$$c_n \leq c_{n+1}$$

for all n . Moreover each consecutive difference is unitarily conjugate to $b - a$, so

$$\tau(c_{n+1} - c_n) = \eta.$$

For $m > n$ we have $c_m - c_n \geq c_{n+1} - c_n$, hence

$$\tau(c_m - c_n) \geq \eta.$$

This is a monotone half-graph of gap η in the terminology of [1, Definition 6.3]. The following argument is the special case of [1, Lemma 6.5]

needed here; we include it for completeness. Suppose that s were the supremum of (c_n) . Then $s - c_n \geq c_{n+1} - c_n$, so

$$\tau(s - c_n) \geq \eta \quad (n \geq 0).$$

Therefore the countable type

$$c_n \leq z \leq s \quad (n \geq 0), \quad \tau(s - z) \geq \eta$$

is finitely satisfiable, by $z = c_n$ for n large. Countable saturation gives an upper bound $z < s$, contradicting the leastness of s . $\square$

Fact 2.8. *If every d_τ -separable type II_1 AW^* -factor is monotone sequentially complete, then every finite AW^* -algebra is monotone complete [1, Corollary 5.3, Theorem 6.10].*

Corollary 2.9. *A tuple (y_1, y_2, y_3) as in Lemma 2.4 gives a type II_1 AW^* -factor which is not monotone sequentially complete. There is also a d_τ -separable such factor.*

Proof. Apply Proposition 2.7 with $\eta = 1/2$. The resulting quasitracial ultrapower is a finite AW^* -factor by Fact 2.6, and the increasing sequence has no supremum. It cannot be type I_n , since those factors are finite dimensional, so it is of type II_1 .

In particular this ultrapower is not monotone complete, so Fact 2.8 gives a d_τ -separable type II_1 AW^* -factor which is not monotone sequentially complete. $\square$

3. BUNDLE-MAP INPUT

In this section, we provide the bundle-theoretic machinery for constructing the configuration in Theorem A. The configuration there is reminiscent of proper infiniteness, but we need it to occur in a finite algebra. We model this behaviour using maps between complex vector bundles.

We begin by showing that generic bundle maps expand fiber subspaces. This uses a modification of the incidence lemma (Lemma 3.1) from the earlier quasitrace construction manuscript. Our proof replaces the incidence variety and the direct covering estimate of the quasitrace manuscript with two standard tools, the stratification of matrices by rank and the parametric transversality theorem. The underlying dimension count is the same.

Lemma 3.1. *Let E and F be complex vector bundles of ranks d and r over a compact smooth manifold X with $\dim_{\mathbb{R}} X \leq 2h$. Fix integers $q \geq 1$, $1 \leq s \leq d$, and $0 \leq t < r$. If*

$$(qs - t)(r - t) - s(d - s) > h,$$

then a generic q -tuple of smooth bundle maps

$$T_1, \dots, T_q : E \longrightarrow F$$

has the following property: for every $x \in X$ and every s -dimensional subspace $S \subseteq E_x$,

$$\dim \sum_{i=1}^q T_i(x)S > t.$$

The conclusion holds simultaneously for any finite set of pairs (s, t) satisfying the displayed inequality. Here generic means that the q -tuples with this property form an open dense subset of the space of smooth q -tuples, with the C^∞ topology.

Proof. Fix Hermitian metrics on E and F . The hypothesis and $r - t > 0$ give $qs - t > 0$.

For $0 \leq k \leq \min\{n_1, n_2\}$, the complex $n_1 \times n_2$ matrices of rank exactly k form a smooth submanifold of complex codimension $(n_1 - k)(n_2 - k)$, invariant under left and right multiplication by invertible matrices. Indeed, near a matrix of rank k we may assume, after permuting rows and columns, that

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

with A_{11} invertible of size k . Then A has rank k exactly when

$$A_{22} - A_{21}A_{11}^{-1}A_{12} = 0,$$

and the left-hand side is a submersion.

Fix one pair (s, t) . Let $\pi : \mathcal{G} \rightarrow X$ be the Grassmann bundle of s -dimensional subspaces of the fibers of E , let $\mathcal{S} \subseteq \pi^*E$ be its tautological subbundle, and put

$$V = \text{Hom}(\mathcal{S} \otimes \mathbb{C}^q, \pi^*F).$$

Then $\mathcal{G}$ is a compact smooth manifold with

$$\dim_{\mathbb{R}} \mathcal{G} \leq 2(h + s(d - s)).$$

For $0 \leq k \leq t$ let $V_k \subseteq V$ consist of the fiber maps of rank exactly k . By the invariance above, V_k is a smooth submanifold of real codimension

$2(qs - k)(r - k)$, and

$$(2) \quad 2(qs - k)(r - k) \geq 2(qs - t)(r - t) > 2(h + s(d - s)) \geq \dim_{\mathbb{R}} \mathcal{G}.$$

A q -tuple $T = (T_1, \dots, T_q)$ determines a section σ_T of V by

$$\sigma_T(x, S) \left(\sum_{i=1}^q \xi_i \otimes e_i \right) = \sum_{i=1}^q T_i(x) \xi_i \quad (\xi_i \in S).$$

Its range is $\sum_i T_i(x)S$, so T has the required property for (s, t) exactly when σ_T misses $V_0 \cup \dots \cup V_t$.

The property is open. If σ_T misses $V_0 \cup \dots \cup V_t$, compactness gives $\varepsilon > 0$ bounding the $(t + 1)$ -st singular value of σ_T from below on $\mathcal{G}$. Singular values are 1-Lipschitz in the operator norm, and

$$\|\sigma_{T'}(g) - \sigma_T(g)\| \leq \sum_{i=1}^q \sup_{x \in X} \|T'_i(x) - T_i(x)\| \quad (g \in \mathcal{G}),$$

so every T' for which the right-hand side is less than ε also has the property.

For density, choose smooth bundle maps $\varphi_1, \dots, \varphi_J: E \rightarrow F$ spanning $\text{Hom}(E_x, F_x)$ at every $x \in X$. For a smooth q -tuple T and $\theta \in \mathbb{C}^{qJ}$ put

$$T_i^\theta = T_i + \sum_{l=1}^J \theta_{il} \varphi_l, \quad \Phi(\theta, g) = \sigma_{T^\theta}(g).$$

For fixed $g = (x, S)$, the map $\theta \mapsto \Phi(\theta, g)$ is affine onto the fiber V_g , since restriction $\text{Hom}(E_x, F_x) \rightarrow \text{Hom}(S, F_x)$ is surjective. As Φ covers the identity of $\mathcal{G}$, it is a submersion $\mathbb{C}^{qJ} \times \mathcal{G} \rightarrow V$, and hence transverse to every V_k . By the parametric transversality theorem [8, Chapter 2, §3], σ_{T^θ} is transverse to every V_k for almost every θ . By (2), transversality means that σ_{T^θ} misses V_k .

For a finite set of pairs (s, t) , the exceptional parameters form a finite union of null sets. Hence T^θ has the property for all the pairs for θ arbitrarily close to 0, and $T^\theta \rightarrow T$ in the C^∞ topology as $\theta \rightarrow 0$. $\square$

We use it only for finitely many pairs (s, t) at each finite test. The second input is a slight extension of the normalization lemma from the quasitrace manuscript. It differs from their Lemma 3.2 in two respects. The expansion factor λ and the normalization constant β may be any reals with $\lambda > \beta > 1$, which we need since both tend to 2 as N grows. More importantly, the normalized maps respect a fixed orthogonal splitting of the target whenever the original maps do. In the quasitrace manuscript, the scaling may mix summands, whereas

here the fresh shifts must keep their range in a prescribed subbundle, since this is how their quasitraces are controlled in Lemma 5.3.

Lemma 3.2. *Let E, F be complex vector bundles over a compact space and let*

$$T_1, \dots, T_q : E \longrightarrow F$$

be continuous bundle maps. Suppose that for every fiber subspace $S \subseteq E_x$,

$$\dim \sum_i T_i(x)S \geq \lambda \dim S$$

for some $\lambda > \beta > 1$. Then there are continuous maps

$$X_1, \dots, X_q : E \longrightarrow F$$

such that

$$\sum_i X_i^* X_i = 1_E, \quad \sum_i X_i X_i^* \leq \beta^{-1} 1_F.$$

If the ranges of the T_i are contained in one of two fixed orthogonal subbundles of F , the corresponding X_i may be chosen with the same range restriction.

Proof. Fix a fiber and, for a positive definite $P \in \text{End}(E_x)$, put

$$D_x(P) = \sum_i T_i(x) P T_i(x)^*,$$

and

$$g_x(P) = \log(\det(1 + D_x(P))) - \beta \log(\det P).$$

We first record uniform coercivity. Put

$$r_s = \lceil \lambda s \rceil, \quad r_0 = 0.$$

Compactness of the rank- s Grassmann bundle and the expansion hypothesis give $\varepsilon > 0$, independent of x and s , such that the r_s -th eigenvalue of $D_x(p)$ is at least ε for every rank- s projection p .

If $t_1 \geq \dots \geq t_d > 0$ are the eigenvalues of P , the min-max principle gives

$$\log(\det(1 + D_x(P))) \geq \sum_{s=1}^d (r_s - r_{s-1}) \log(1 + \varepsilon t_s).$$

Since $r_s - \lambda s \geq 0$, summation by parts against the decreasing nonnegative sequence $\log(1 + \varepsilon t_s)$ yields

$$\log(\det(1 + D_x(P))) \geq \lambda \sum_{s=1}^d \log(1 + \varepsilon t_s).$$

The scalar function

$$\lambda \log(1 + \varepsilon t) - \beta \log(t)$$

tends to $+\infty$ as $t \rightarrow 0$ or $t \rightarrow \infty$. Hence the sublevel sets of g_x are uniformly compact.

Along a geodesic

$$P(u) = P_0^{1/2} e^{uH} P_0^{1/2}$$

in the positive cone, $\det(1 + D_x(P(u)))$ is a positive sum of exponentials in u . It contains the constant term 1, and if $H \neq 0$ it contains a nonconstant exponential with positive coefficient because the common kernel of the $T_i(x)$ is zero. Thus the first term of $g_x(P(u))$ is strictly convex, while $\log \det P(u)$ is affine. There is a unique minimizer P_x , and uniform coercivity plus uniqueness shows that $x \mapsto P_x$ is continuous.

Put

$$Q_x = 1 + D_x(P_x).$$

Differentiating at the minimum gives

$$\sum_i T_i(x)^* Q_x^{-1} T_i(x) = \beta P_x^{-1}.$$

Define

$$X_i = \beta^{-1/2} Q^{-1/2} T_i P^{1/2}.$$

Then

$$\sum_i X_i^* X_i = 1_E$$

and

$$\sum_i X_i X_i^* = \beta^{-1} Q^{-1/2} D(P) Q^{-1/2} = \beta^{-1} (1 - Q^{-1}) \leq \beta^{-1} 1_F.$$

If the ranges split between two fixed orthogonal subbundles, then $D(P)$ and Q are block diagonal for that decomposition, so $Q^{-1/2}$ preserves the two summands. $\square$

We now show that blockwise expansion gives global expansion. The difficulty is that an arbitrary subspace S need not decompose as the direct sum of its intersections with the blocks. The proof works by filtering S through successive partial sums and measuring the dimensions contributed in the next block via a projected subspace.

Lemma 3.3. *Let $V = \bigoplus_{k=0}^n V_k$ and $W = \bigoplus_{k=0}^n W_k$ be orthogonal direct sums of finite-dimensional Hilbert spaces, and let $T_1, \dots, T_q: V \rightarrow W$ be block diagonal with blocks $T_l^{(k)}: V_k \rightarrow W_k$. Suppose that*

$$\dim \sum_{l=1}^q T_l^{(k)} G \geq \lambda \dim G$$

for every k and every subspace $G \subseteq V_k$ with $\dim G \leq \kappa$. Then

$$\dim \sum_{l=1}^q T_l S \geq \lambda \dim S$$

for every subspace $S \subseteq V$ with $\dim S \leq \kappa$.

Proof. Let π_k and ϖ_k be the orthogonal projections onto V_k and W_k . Put

$$S_k = S \cap (V_0 \oplus \dots \oplus V_k), \quad G_k = \pi_k(S_k), \quad I_k = \sum_l T_l S_k,$$

and $S_{-1} = I_{-1} = 0$. The restriction of π_k to S_k has kernel S_{k-1} , so

$$\dim S = \sum_{k=0}^n \dim G_k.$$

Since the T_l are block diagonal, $I_{k-1} \subseteq \ker \varpi_k$ and $\varpi_k(I_k) = \sum_l T_l^{(k)} G_k$. Hence

$$\dim I_k \geq \dim I_{k-1} + \dim \sum_l T_l^{(k)} G_k \geq \dim I_{k-1} + \lambda \dim G_k,$$

because $\dim G_k \leq \dim S \leq \kappa$. Summing over k gives the claim. $\square$

4. THE FINITE BUNDLE TESTS

In this section, we combine the geometric machinery of the last with a topological obstruction. We need to balance two competing properties: that there are enough maps to produce strong compression inequalities, but not enough topological freedom to produce isometries that would witness proper infiniteness.

Fix an integer $N \geq 4$ and set

$$\rho = 1 - \frac{1}{N}, \quad \lambda = 2 - \frac{2}{N}, \quad \beta = 2 - \frac{3}{N}, \quad \gamma_N = \beta^{-1}.$$

Thus

$$\lambda > \beta > 1, \quad \gamma_N \longrightarrow \frac{1}{2}.$$

For integers $m, R \geq 1$ put

$$s_0 = Nm, \quad d_0 = 16N^2mN^{R+1}, \quad d_j = 4^j d_0.$$

For $0 \leq j \leq R$ define

$$a_{0,j} = \rho^j d_0,$$

and, for $1 \leq i \leq j \leq R$,

$$a_{i,j} = (4 - \rho)d_{i-1}\rho^{j-i}.$$

The choice of d_0 makes all $a_{i,j}$ positive integers. The recurrence

$$\sum_{i=0}^j a_{i,j} = d_j$$

follows by induction. For $j < R$, every $a_{i,j}$ is divisible by N and

$$a_{i,j} \geq 16Ns_0.$$

For $1 \leq i \leq R$ set

$$K_i = m \sum_{j=i}^R a_{i,j}$$

and let

$$Y = \prod_{i=1}^R \mathbb{CP}^{K_i}.$$

Write L_i for the pullback of the hyperplane line bundle from the i -th factor and put $L_0 = 1$. Define

$$E_j = \bigoplus_{i=0}^j L_i^{\oplus a_{i,j}}.$$

Then $\text{rank } E_j = d_j$. For $j < R$ there is a canonical decomposition

$$E_{j+1} = O_{j+1} \oplus F_{j+1},$$

where

$$O_{j+1} = \bigoplus_{i=0}^j L_i^{\oplus \rho a_{i,j}}, \quad F_{j+1} = L_{j+1}^{\oplus (4-\rho)d_j}.$$

We refer to these as the old and fresh summands respectively. Each kind of summand does a different job. The old maps will survive the quasitracial ultraproduct. The fresh maps enforce expansion at each finite stage but will disappear in that ultraproduct.

Proposition 4.1. *For every $j < R$ there are eleven continuous bundle maps*

$$X_1^{(j)}, \dots, X_{11}^{(j)}: E_j \longrightarrow E_{j+1}$$

such that

$$\sum_{k=1}^{11} (X_k^{(j)})^* X_k^{(j)} = 1_{E_j}, \quad \sum_{k=1}^{11} X_k^{(j)} (X_k^{(j)})^* \leq \gamma_N 1_{E_{j+1}},$$

the first three maps have range in O_{j+1} , and the remaining eight have range in F_{j+1} .

Proof. We first construct three maps into the old summand. Fix a multiplicity block $L_i^{\oplus D}$ in E_j , so the corresponding old block has rank ρD . Choose three generic scalar maps

$$\mathbb{C}^D \longrightarrow \mathbb{C}^{\rho D}.$$

For $1 \leq s \leq s_0$ put $t = \lfloor \lambda s \rfloor$. Since $t \leq \lambda s$ and $\rho = 1 - 1/N$, the incidence expression of Lemma 3.1, with base a point, is bounded below by

$$(3s - t)(\rho D - t) - s(D - s) \geq ((3 - \lambda)\rho - 1)sD - \frac{5}{4}s^2 \geq \frac{sD}{2N} - \frac{5}{4}s^2 > 0.$$

Here we used

$$(3 - \lambda)\rho - 1 = \frac{1}{N} - \frac{2}{N^2} \geq \frac{1}{2N}, \quad (3 - \lambda)\lambda - 1 = 1 + \frac{2}{N} - \frac{4}{N^2} \leq \frac{5}{4},$$

and the last inequality uses $D \geq 16Ns_0$ and $s \leq s_0$. Thus the three maps may be chosen so that the collective image of every s -dimensional subspace, $s \leq s_0$, has dimension greater than $\lfloor \lambda s \rfloor$, hence at least λs . Tensor with L_i and take block sums over i . On each fiber, Lemma 3.3, applied to the line-bundle blocks, shows that these three old maps expand every fiber subspace of dimension at most s_0 by a factor at least λ .

We next construct eight maps into the fresh summand. It is enough to work over

$$Y_{j+1} = \prod_{i=1}^{j+1} \mathbb{CP}^{K_i},$$

since both E_j and F_{j+1} are pulled back from this factor. Its complex dimension satisfies

$$h_{j+1} = \sum_{i=1}^{j+1} K_i \leq Nm \sum_{i=1}^{j+1} (4 - \rho)d_{i-1} < \frac{4(4 - \rho)}{3} Nmd_j < 5Nmd_j.$$

Fix $s > s_0$, set $t = \lfloor \lambda s \rfloor \leq 2s$, $d = d_j$, and

$$r = (4 - \rho)d \geq 3d.$$

For eight maps the incidence expression satisfies

$$\begin{aligned} (8s - t)(r - t) - s(d - s) &\geq 6s(3d - 2s) - s(d - s) \\ &= 17sd - 11s^2 \\ &\geq 6sd \\ &> h_{j+1}. \end{aligned}$$

Since $s \leq d$, the third inequality is immediate, and the last uses $s > s_0 = Nm$. Lemma 3.1, applied simultaneously to the finitely many values $s_0 < s \leq d$, gives eight fresh maps which expand every subspace of dimension greater than s_0 by a factor at least λ .

Together the three old and eight fresh maps expand every fiber subspace by a factor at least λ . Lemma 3.2 gives the required normalized maps, and its final assertion preserves the old/fresh range division. $\square$

We now record the obstruction which will prevent stable proper infiniteness.

Lemma 4.2. *Put*

$$G = \left(\bigoplus_{j=0}^R E_j \right)^{\oplus m}.$$

Then G has no trivial subbundle of rank $(N + 1)md_0$.

Proof. Let

$$h_0 = m \sum_{j=0}^R a_{0,j}.$$

Since $\sum_{j=0}^R \rho^j < N$, we have

$$h_0 < Nmd_0.$$

By the definition of the K_i ,

$$G \cong \mathbb{C}^{h_0} \oplus \bigoplus_{i=1}^R L_i^{\oplus K_i}.$$

Put

$$p = \sum_{i=1}^R K_i.$$

Then

$$c_p(G) = \prod_{i=1}^R c_1(L_i)^{K_i}.$$

This is the nonzero top product in

$$H^{2p}\left(\prod_{i=1}^R \mathbb{CP}^{K_i}; \mathbb{Z}\right).$$

If G contained a trivial subbundle of rank $k > h_0$, its orthogonal complement would have rank

$$h_0 + p - k < p.$$

The degree- p Chern class of G would then vanish, a contradiction. Since

$$(N+1)md_0 > h_0,$$

the stated trivial subbundle cannot exist. $\square$

5. THE FINITELY GENERATED TEST ALGEBRAS

We now turn our finite bundle tests into actual C^* -algebras. The test bundles are assembled into Hilbert bundles carrying thirteen forward shifts. Two shifts operate on a tail where bundles double, three operate in the old portions of the finite stages, and eight operate in the fresh portions. The resulting algebra B_N is generated by these shifts together with finitely many additional operators controlling the old/fresh decomposition.

Continue to fix $N \geq 4$. For each pair (m, R) and its bundle system above, extend the sequence beyond level R by

$$E_{j+1} = E_j \oplus E_j \quad (j \geq R).$$

Let

$$H_{m,R} = \bigoplus_{j \geq 0} E_j$$

be the resulting Hilbert bundle over Y . We work with bounded continuous adjointable bundle maps on these Hilbert bundles, and finally take the product over all (m, R) .

Define thirteen forward shifts. The first two vanish on levels $j < R$ and, for $j \geq R$, are $2^{-1/2}$ times the two canonical inclusions

$$E_j \longrightarrow E_j \oplus E_j = E_{j+1}.$$

The remaining eleven act on E_j , for $j < R$, by the maps from Proposition 4.1, and vanish on the tail. Denote the resulting global shifts by

$$s_1, \dots, s_{13}.$$

They satisfy

$$\sum_{k=1}^{13} s_k^* s_k = 1, \quad \sum_{k=1}^{13} s_k s_k^* \leq \gamma_N 1.$$

Let h be the projection onto the levels $0, \dots, R-1$ in each test and put $q = 1 - h$. For $j < R$, every multiplicity $a_{i,j}$ is divisible by N . Divide the copies in every line-bundle multiplicity into N equal groups. This gives block-diagonal matrix units

$$(e_{rs})_{r,s=1}^N, \quad \sum_{r=1}^N e_{rr} = h.$$

Put

$$p = \sum_{r=1}^{N-1} e_{rr}.$$

The selected ρ -fraction of the copies at level j is bundle-isomorphic to O_{j+1} . Consequently there is a continuous propagation-one partial isometry v with

$$v^* v = p, \quad v v^* = O := \bigoplus_{j=0}^{R-1} 1_{O_{j+1}}.$$

Let B_N be the unital C^* -subalgebra of the product over all tests (m, R) generated by

$$s_1, \dots, s_{13}, \quad h, \quad (e_{rs})_{r,s=1}^N, \quad v.$$

This is a finitely generated unital C^* -algebra. Every displayed generator and its adjoint has propagation at most one.

We can now show that each B_N has no properly infinite matrix amplification. The proof works by approximating any witness to proper infiniteness by noncommutative polynomials and using these to build a map forbidden by Lemma 4.2.

Lemma 5.1. *No matrix algebra over B_N is properly infinite.*

Proof. Suppose that $M_m(B_N)$ is properly infinite for some m . Iterating the two-isometry characterization of proper infiniteness gives $N + 1$ isometries with mutually orthogonal ranges. Regard them as a row

$$W \in M_{m,(N+1)m}(B_N), \quad W^* W = 1.$$

Approximate W sufficiently closely by a row P whose entries are non-commutative polynomials in the displayed generators and their adjoints. We may arrange

$$P^*P \geq \frac{1}{2}1.$$

Let R bound the word lengths of these polynomials, increasing it to at least one.

Evaluate P at the test indexed by this same pair (m, R) . Since every generator has propagation at most one, the restriction of P to level zero has range in the first $R + 1$ levels. Thus it is a continuous bundle map

$$E_0^{\oplus(N+1)m} \longrightarrow \left(\bigoplus_{j=0}^R E_j \right)^{\oplus m}.$$

The inequality $P^*P \geq \frac{1}{2}1$ makes this map fiberwise injective with continuous image subbundle. Its source is trivial of rank

$$(N + 1)md_0,$$

contradicting Lemma 4.2. $\square$

Fact 5.2. *A unital C^* -algebra admits a normalized 2-quasitrace if and only if it is not stably properly infinite [7, 11, 4, 5]; see [13, Theorem 2.3].*

Hence B_N admits normalized 2-quasitraces. We now show that such a quasitrace makes the fresh region small.

Lemma 5.3. *Let τ be any normalized 2-quasitrace on B_N . Then*

$$\tau(q) = 0, \quad \tau(h) = 1, \quad \tau(p) = \tau(O) = \rho, \quad \tau(1 - O) = \frac{1}{N}.$$

If $f_1, \dots, f_8$ denote the eight fresh shifts among $s_3, \dots, s_{13}$, then

$$\tau(f_j^* f_j) \leq \frac{\gamma_N}{N} \quad (j = 1, \dots, 8).$$

Proof. The first two tail shifts, multiplied by $\sqrt{2}$, are partial isometries with initial projection q and mutually orthogonal range projections below q . Quasitracial invariance under Murray–von Neumann equivalence and additivity on orthogonal projections give

$$2\tau(q) \leq \tau(q),$$

so $\tau(q) = 0$ and $\tau(h) = 1$.

The matrix units show that

$$\tau(e_{11}) = \cdots = \tau(e_{NN}) = \frac{1}{N},$$

whence

$$\tau(p) = \frac{N-1}{N} = \rho.$$

Since $v^*v = p$ and $vv^* = O$,

$$\tau(O) = \rho,$$

and therefore $\tau(1 - O) = 1/N$.

Every fresh shift has range in $1 - O$, while the row estimate gives

$$f_j f_j^* \leq \gamma_N (1 - O).$$

Hence

$$\tau(f_j^* f_j) = \tau(f_j f_j^*) \leq \gamma_N \tau(1 - O) = \frac{\gamma_N}{N}. \quad \square$$

Fact 5.4. *Let τ be an extremal normalized 2-quasitrace on a unital C^* -algebra B . Dividing out the null ideal of τ and taking Haagerup's quasitracial AW^* -completion gives a finite AW^* -factor whose canonical quasitrace extends τ [10, Proposition 4.6]; see also [13, Section 2].*

Choose an extremal point τ_N of the compact convex set of normalized 2-quasitraces on B_N , and let M_N be the finite AW^* -factor given by Fact 5.4. Extremality passes to the null quotient, since any convex decomposition there pulls back to one on B_N . All of the displayed algebraic relations and the quasitrace identities of Lemma 5.3 pass to M_N .

The two tail shifts vanish in M_N , because their initial projection q belongs to the null ideal. Let

$$z_{1,N}, \quad z_{2,N}, \quad z_{3,N}$$

be the three old shifts and let

$$f_{1,N}, \dots, f_{8,N}$$

be the eight fresh shifts. Then

$$\sum_{i=1}^3 z_{i,N}^* z_{i,N} + \sum_{j=1}^8 f_{j,N}^* f_{j,N} = 1,$$

and

$$\sum_{i=1}^3 z_{i,N} z_{i,N}^* \leq \gamma_N 1.$$

Moreover,

$$\tau_N(f_{j,N}^* f_{j,N}) \leq \frac{\gamma_N}{N}.$$

We can now show failure of Kaplansky's quasitrace conjecture, which was previously claimed in the quasitrace manuscript.

Proposition 5.5. *The quasitrace τ_N on B_N is not a trace, and M_N is a type II_1 AW^* -factor which is not a von Neumann algebra.*

Proof. If τ_N were additive on B_N , then

$$1 = \sum_{k=1}^{13} \tau_N(s_k^* s_k) = \sum_{k=1}^{13} \tau_N(s_k s_k^*) = \tau_N\left(\sum_{k=1}^{13} s_k s_k^*\right) \leq \gamma_N < 1.$$

The same computation applies to the canonical quasitrace of M_N . The normalized quasitrace of a matrix algebra is its trace, so M_N is of type II_1 . An AW^* -factor of type II_1 is a von Neumann algebra if and only if its quasitrace is a trace [13, Section 2], so M_N is not a von Neumann algebra. $\square$

Remark 5.6. In the notation of [13, Remark 4.7], Lemma 5.1 gives $\mu(B_N) = \infty$. The 13^4 products $s_K = s_{k_1} s_{k_2} s_{k_3} s_{k_4}$ of the shifts satisfy

$$\sum_K s_K^* s_K = 1, \quad \sum_K s_K s_K^* \leq \gamma_N^4 1 \leq \frac{1}{2} 1,$$

since $\gamma_N \leq 4/5$. Hence $\nu(B_N) \leq 13^4$ for every $N \geq 4$.

6. PASSING TO THE CONFIGURATION

Let N run through the integers at least 4, and fix a nonprincipal ultrafilter $\mathcal{V}$ on this set. Form the quasitracial ultraproduct

$$M = \prod_{N \rightarrow \mathcal{V}} (M_N, \tau_N).$$

By [1, Corollary 4.13], M is a finite AW^* -factor.

Theorem A. *The factor M contains elements y_1, y_2, y_3 satisfying*

$$\sum_{i=1}^3 y_i^* y_i = 1, \quad \sum_{i=1}^3 y_i y_i^* \leq \frac{1}{2} 1.$$

Proof. For each fixed $j \leq 8$,

$$d_\tau(f_{j,N}, 0)^3 = \tau_N(f_{j,N}^* f_{j,N}) \leq \frac{\gamma_N}{N} \longrightarrow 0.$$

Hence the fresh shifts represent zero in the quasitracial ultraproduct. Put

$$y_i = [z_{i,N}] \quad (i = 1, 2, 3).$$

The exact column relation in each M_N therefore gives

$$\sum_{i=1}^3 y_i^* y_i = 1.$$

For the row relation,

$$\gamma_N 1 - \sum_{i=1}^3 z_{i,N} z_{i,N}^* \geq 0.$$

Since $\gamma_N \rightarrow 1/2$, passage to the ultraproduct gives

$$\frac{1}{2} 1 - \sum_{i=1}^3 y_i y_i^* \geq 0. \quad \square$$

Theorem B. *There is a type II_1 AW^* -factor which is not monotone sequentially complete. Moreover, there is such a factor which is d_τ -separable.*

Proof. Apply Corollary 2.9 to the tuple of Theorem A. $\square$

The constants are not optimized. The role of the parameter N is to make the old fraction ρ tend to one, the normalization constant γ_N tend to $1/2$, and the complementary fresh range have quasitrace $1/N$. The fixed number eight of fresh maps is then essential: their total contribution vanishes in the ultraproduct.

7. FURTHER CONSEQUENCES

We now list some further consequences.

Corollary 7.1. *There is a unital stably finite C^* -algebra generated by three elements and admitting no tracial state. Moreover, for the universal unital C^* -algebra*

$$\mathcal{D}_3 = C^* \left\langle x_1, x_2, x_3 \left| \sum_{i=1}^3 x_i^* x_i = 1, \quad \sum_{i=1}^3 x_i x_i^* \leq \frac{1}{2} 1 \right. \right\rangle,$$

no matrix amplification of $\mathcal{D}_3$ is properly infinite. The number three is minimal among stably finite unital C^ -algebras admitting this configuration.*

Proof. Let M and y_1, y_2, y_3 be as in Theorem A, and set $D = C^*(1, y_1, y_2, y_3) \subseteq M$. Each matrix amplification of the finite AW*-factor M is finite, and hence so is each matrix amplification of D ; thus D is stably finite. If D had a tracial state φ , the relations would give

$$1 = \sum_{i=1}^3 \varphi(y_i^* y_i) = \sum_{i=1}^3 \varphi(y_i y_i^*) \leq \frac{1}{2},$$

a contradiction. The universal property yields a unital surjection $\mathcal{D}_3 \rightarrow D$. Proper infiniteness of any matrix amplification would pass to this quotient, which proves the second assertion.

For minimality, suppose x_1, x_2 satisfy the same relations with only two terms. The row bound implies $x_i^* x_i \leq \frac{1}{2}1$, whereas $x_1^* x_1 + x_2^* x_2 = 1$; hence $x_1^* x_1 = x_2^* x_2 = \frac{1}{2}1$. The elements $\sqrt{2}x_i$ are isometries, and the row inequality implies their range projections are orthogonal. The algebra is therefore properly infinite and cannot be stably finite. $\square$

Corollary 7.2. *There is a type II₁ AW*-factor P such that*

P is not a W-algebra, every masa of P is a W*-algebra.*

Moreover, P can be chosen d_τ -separable, in which case every masa of P has separable predual. Such a factor has no faithful state. Consequently the finite case of Gow's intrinsic W-Pedersen conjecture [9, Conjecture 3.15] has a negative answer.*

Proof. Choose the d_τ -separable factor P from Theorem B. It is not monotone sequentially complete, so it is not a W*-algebra. Every masa $C \subseteq P$ is a W*-algebra by the finite-factor masa theorem [9, Proposition 2.32] (equivalently, by the completeness of its unit ball for the metric induced by the faithful restriction of τ). Because P is d_τ -separable, so is the unit ball of C . On C , the metric $d_\tau(x, y) = \|x - y\|_{2, \tau}^{2/3}$ is a power of the usual L^2 -metric. Thus $L^2(C, \tau)$ is separable; by the density of bounded functions in $L^1(C, \tau)$, the predual $C_* \cong L^1(C, \tau)$ is separable. Finally, a finite AW*-factor with a faithful state is a W*-factor by Wright's theorem [15]. Hence P has no faithful state. $\square$

Corollary 7.3. *There is a finite real AW*-algebra R which is not monotone sequentially complete and admits no center-valued trace. For this R , there is neither a strictly positive real-linear center-module map $R \rightarrow Z(R)$ nor a faithful norm-one real-linear projection $R \rightarrow Z(R)$.*

Proof. Regard the non-W*-type II₁ factor P of Corollary 7.2 as a real C*-algebra R . The real and complex algebras have the same projections

and self-adjoint order, so R is a finite real AW^* -algebra and inherits the increasing sequence without a supremum. Its real center is $Z(R) = \mathbb{C}1$. A center-valued trace $T : R \rightarrow Z(R)$ would be $Z(R)$ -linear and hence complex-linear, so it would yield a faithful complex trace on P , contradicting its nontracial canonical quasitrace. The remaining nonexistence assertions follow from the equivalences in [1, Theorem 7.34]. $\square$

REFERENCES

- [1] J. Arulseelan, *Model theory of quasitracial AW^* -algebras and the absoluteness of monotone completeness*, preprint.
- [2] J. Arulseelan and J. E. Hanson, *Every AW^* -algebra is normal*, preprint, 2026.
- [3] S. K. Berberian, *Baer $*$ -rings*, Die Grundlehren der mathematischen Wissenschaften, vol. 195, Springer-Verlag, New York–Berlin, 1972.
- [4] B. Blackadar and D. Handelman, *Dimension functions and traces on C^* -algebras*, J. Funct. Anal. **45** (1982), 297–340.
- [5] B. Blackadar and M. Rørdam, *Extending states on preordered semigroups and the existence of quasitraces on C^* -algebras*, J. Algebra **152** (1992), 240–247.
- [6] E. Christensen and G. K. Pedersen, *Properly infinite AW^* -algebras are monotone sequentially complete*, Bull. London Math. Soc. **16** (1984), 407–410.
- [7] J. Cuntz, *Dimension functions on simple C^* -algebras*, Math. Ann. **233** (1978), 145–153.
- [8] V. Guillemin and A. Pollack, *Differential topology*, Prentice-Hall, Englewood Cliffs, NJ, 1974.
- [9] A. Gow, *On the quasitrace problem and a characterization of W^* -algebras*, preprint, arXiv:2601.04431, 2026.
- [10] U. Haagerup, *Quasitraces on exact C^* -algebras are traces*, C. R. Math. Rep. Acad. Sci. Canada **36** (2014), no. 2–3, 67–92.
- [11] D. Handelman, *Homomorphisms of C^* -algebras to finite AW^* -algebras*, Michigan Math. J. **28** (1981), 229–240.
- [12] I. Kaplansky, *Projections in Banach algebras*, Ann. of Math. (2) **53** (1951), 235–249.
- [13] H. O. Milhøj and M. Rørdam, *Around traces and quasitraces*, arXiv:2309.17412, 2023.
- [14] K. Saitô and J. D. M. Wright, *All AW^* -factors are normal*, J. London Math. Soc. (2) **44** (1991), 143–154.
- [15] J. D. M. Wright, *On AW^* -algebras of finite type*, J. London Math. Soc. (2) **12** (1976), 431–439.

DEPARTMENT OF MATHEMATICS, IOWA STATE UNIVERSITY, 396 CARVER HALL,
411 MORRILL ROAD, AMES, IA 50011, USA

Email address: jananan@iastate.edu

URL: <https://sites.google.com/view/jananan-arulseelan>

COUNTEREXAMPLES TO THE MONOTONE COMPLETENESS CONJECTURE⁵

DEPARTMENT OF MATHEMATICS, IOWA STATE UNIVERSITY, 396 CARVER HALL,
411 MORRILL ROAD, AMES, IA 50011, USA

Email address: jameseh@iastate.edu

URL: <https://james-hanson.github.io/>